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Jameco Part Number 46906FSC

## DM74LS169A

### Synchronous 4-Bit Up/Down Binary Counter

#### General Description

This synchronous presettable counter features an internal carry look-ahead for cascading in high-speed counting applications. Synchronous operation is provided by having all flip-flops clocked simultaneously, so that the outputs all change at the same time when so instructed by the count-enable inputs and internal gating. This mode of operation helps eliminate the output counting spikes that are normally associated with asynchronous (ripple clock) counters. A buffered clock input triggers the four master-slave flip-flops on the rising edge of the clock waveform.

This counter is fully programmable; that is, the outputs may each be preset either HIGH or LOW. The load input circuitry allows loading with the carry-enable output of cascaded counters. As loading is synchronous, setting up a low level at the load input disables the counter and causes the outputs to agree with the data inputs after the next clock pulse.

The carry look-ahead circuitry permits cascading counters for n-bit synchronous applications without additional gating. Both count-enable inputs ( $\bar{P}$  and  $\bar{T}$ ) must be LOW to count. The direction of the count is determined by the level of the UP/DOWN input. When the input is HIGH, the counter counts UP; when LOW, it counts DOWN. Input  $\bar{T}$  is fed forward to enable the carry outputs. The carry output thus

enabled will produce a low-level output pulse with a duration approximately equal to the high portion of the  $Q_A$  output when counting UP, and approximately equal to the low portion of the  $Q_A$  output when counting DOWN. This low-level overflow carry pulse can be used to enable successively cascaded stages. Transitions at the enable  $\bar{P}$  or  $\bar{T}$  inputs are allowed regardless of the level of the clock input. All inputs are diode clamped to minimize transmission-line effects, thereby simplifying system design.

This counter features a fully independent clock circuit. Changes at control inputs (enable  $\bar{P}$ , enable  $\bar{T}$ , load, UP/DOWN), which modify the operating mode, have no effect until clocking occurs. The function of the counter (whether enabled, disabled, loading, or counting) will be dictated solely by the conditions meeting the stable setup and hold times.

#### Features

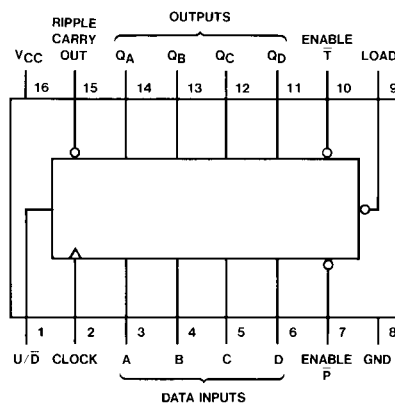
- Fully synchronous operation for counting and programming.
- Internal look-ahead for fast counting.
- Carry output for n-bit cascading.
- Fully independent clock circuit

#### Ordering Code:

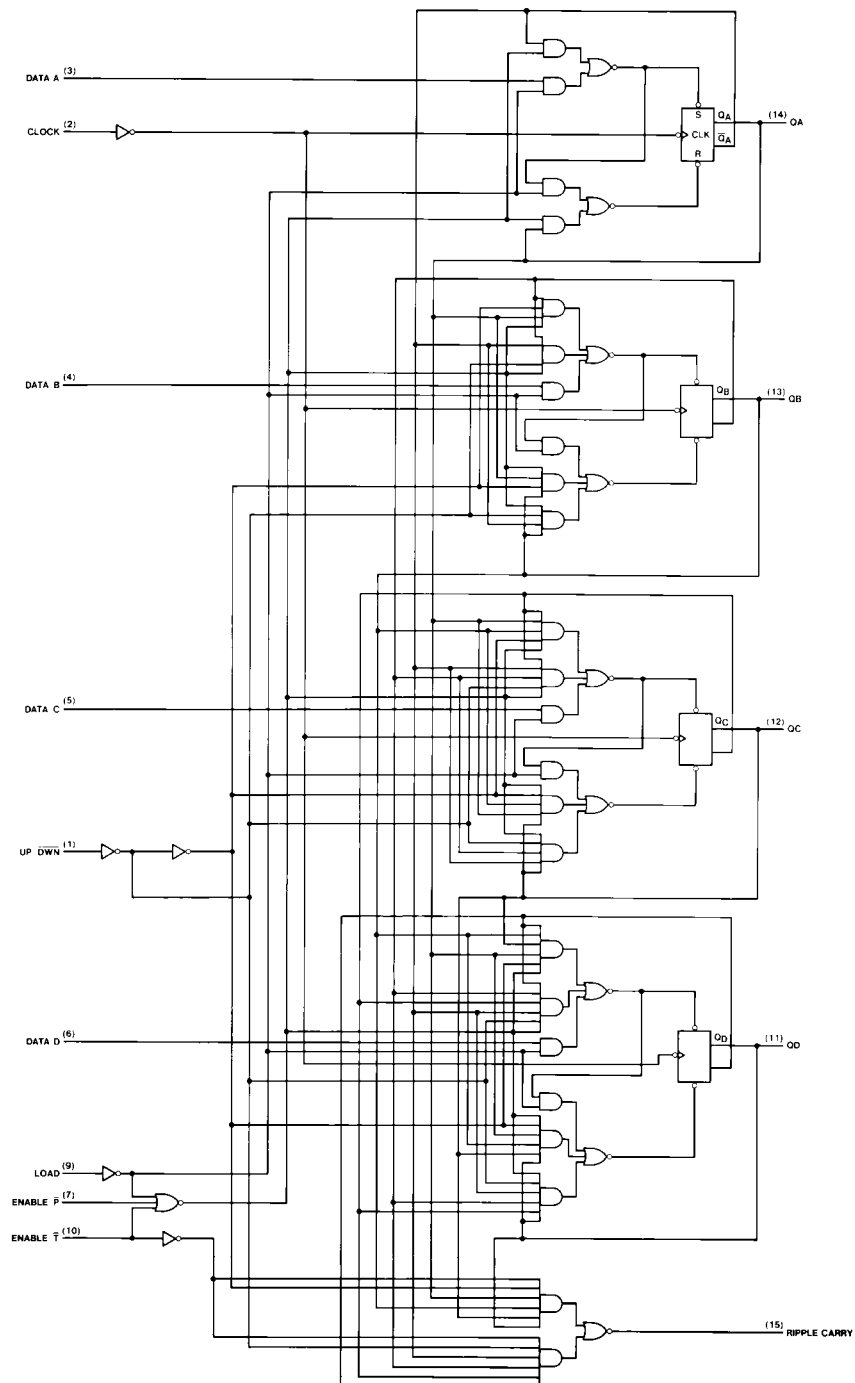
| Order Number | Package Number | Package Description                                                         |
|--------------|----------------|-----------------------------------------------------------------------------|
| DM74LS169AM  | M16A           | 16-Lead Small Outline Integrated Circuit (SOIC), JEDEC MS-012, 0.150 Narrow |
| DM74LS169AN  | N16E           | 16-Lead Plastic Dual-In-Line Package (PDIP), JEDEC MS-001, 0.300 Wide       |

Devices also available in Tape and Reel. Specify by appending the suffix letter "X" to the ordering code.

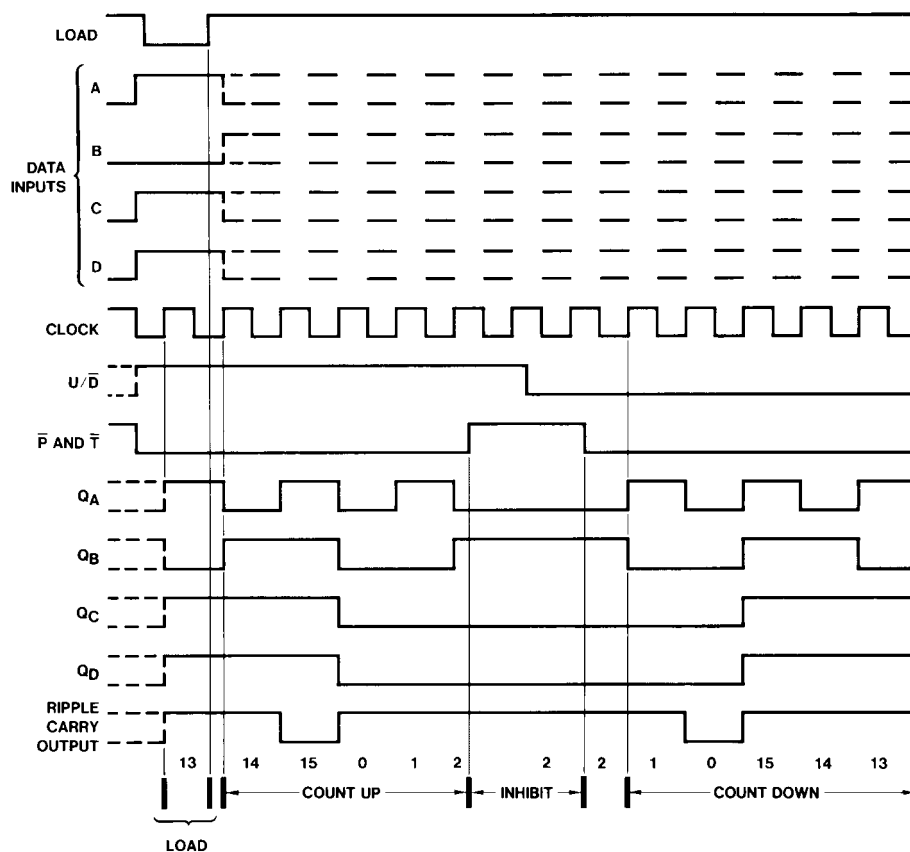
#### Connection Diagram



## Logic Diagram



## Timing Diagram



Typical Load, Count, and Inhibit Sequences

**Absolute Maximum Ratings**(Note 1)

|                                      |                 |
|--------------------------------------|-----------------|
| Supply Voltage                       | 7V              |
| Input Voltage                        | 7V              |
| Operating Free Air Temperature Range | 0°C to +70°C    |
| Storage Temperature Range            | –65°C to +150°C |

**Note 1:** The "Absolute Maximum Ratings" are those values beyond which the safety of the device cannot be guaranteed. The device should not be operated at these limits. The parametric values defined in the Electrical Characteristics tables are not guaranteed at the absolute maximum ratings. The "Recommended Operating Conditions" table will define the conditions for actual device operation.

**Recommended Operating Conditions**

| Symbol    | Parameter                      | Min                              | Nom | Max  | Units |
|-----------|--------------------------------|----------------------------------|-----|------|-------|
| $V_{CC}$  | Supply Voltage                 | 4.75                             | 5   | 5.25 | V     |
| $V_{IH}$  | HIGH Level Input Voltage       | 2                                |     |      | V     |
| $V_{IL}$  | LOW Level Input Voltage        |                                  |     | 0.8  | V     |
| $I_{OH}$  | HIGH Level Output Current      |                                  |     | –0.4 | mA    |
| $I_{OL}$  | LOW Level Output Current       |                                  |     | 8    | mA    |
| $f_{CLK}$ | Clock Frequency (Note 2)       | 0                                |     | 25   | MHz   |
|           | Clock Frequency (Note 3)       | 0                                |     | 20   | MHz   |
| $t_W$     | Clock Pulse Width (Note 4)     | 25                               |     |      | ns    |
| $t_{SU}$  | Setup Time<br>(Note 4)         | Data                             | 20  |      | ns    |
|           |                                | Enable<br>$\bar{T}$ or $\bar{P}$ | 20  |      |       |
|           |                                | Load                             | 25  |      |       |
|           |                                | U/D                              | 30  |      |       |
| $t_H$     | Hold Time (Note 4)             | 0                                |     |      | ns    |
| $T_A$     | Free Air Operating Temperature | 0                                |     | 70   | °C    |

**Note 2:**  $C_L = 15$  pF,  $R_L = 2$  k $\Omega$ ,  $T_A = 25^\circ\text{C}$  and  $V_{CC} = 5\text{V}$ .

**Note 3:**  $C_L = 50$  pF,  $R_L = 2$  k $\Omega$ ,  $T_A = 25^\circ\text{C}$  and  $V_{CC} = 5\text{V}$ .

**Note 4:**  $T_A = 25^\circ\text{C}$  and  $V_{CC} = 5\text{V}$ .

**Electrical Characteristics**

over recommended operating free air temperature range (unless otherwise noted)

| Symbol   | Parameter                            | Conditions                                    | Min | Typ<br>(Note 5) | Max  | Units         |
|----------|--------------------------------------|-----------------------------------------------|-----|-----------------|------|---------------|
| $V_I$    | Input Clamp Voltage                  | $V_{CC} = \text{Min}$ , $I_I = -18$ mA        |     |                 | –1.5 | V             |
| $V_{OH}$ | HIGH Level                           | $V_{CC} = \text{Min}$ , $I_{OH} = \text{Max}$ | 2.7 | 3.4             |      | V             |
|          | Output Voltage                       | $V_{IL} = \text{Max}$ , $V_{IH} = \text{Min}$ |     |                 |      |               |
| $V_{OL}$ | LOW Level<br>Output Voltage          | $V_{CC} = \text{Min}$ , $I_{OL} = \text{Max}$ |     | 0.35            | 0.5  | V             |
|          |                                      | $V_{IL} = \text{Max}$ , $V_{IH} = \text{Min}$ |     |                 |      |               |
| $I_I$    | Input Current @ Max<br>Input Voltage | $V_{CC} = \text{Max}$                         |     |                 | 0.2  | mA            |
|          |                                      | $V_I = 7\text{V}$                             |     |                 |      |               |
| $I_{IH}$ | HIGH Level<br>Input Current          | $V_{CC} = \text{Max}$                         |     |                 | 40   | $\mu\text{A}$ |
|          |                                      | $V_I = 2.7\text{V}$                           |     |                 |      |               |
| $I_{IL}$ | LOW Level<br>Input Current           | $V_{CC} = \text{Max}$                         |     |                 | –0.8 | mA            |
|          |                                      | $V_I = 0.4\text{V}$                           |     |                 |      |               |
| $I_{OS}$ | Short Circuit Output Current         | $V_{CC} = \text{Max}$ (Note 6)                | –20 |                 | –100 | mA            |
| $I_{CC}$ | Supply Current                       | $V_{CC} = \text{Max}$ (Note 7)                |     | 20              | 34   | mA            |

**Note 5:** All typicals are at  $V_{CC} = 5\text{V}$  and  $T_A = 25^\circ\text{C}$ .

**Note 6:** Not more than one output should be shorted at a time, and the duration should not exceed one second.

**Note 7:**  $I_{CC}$  is measured after a momentary 4.5V, then ground, is applied to the CLOCK with all other inputs grounded and all the outputs OPEN.

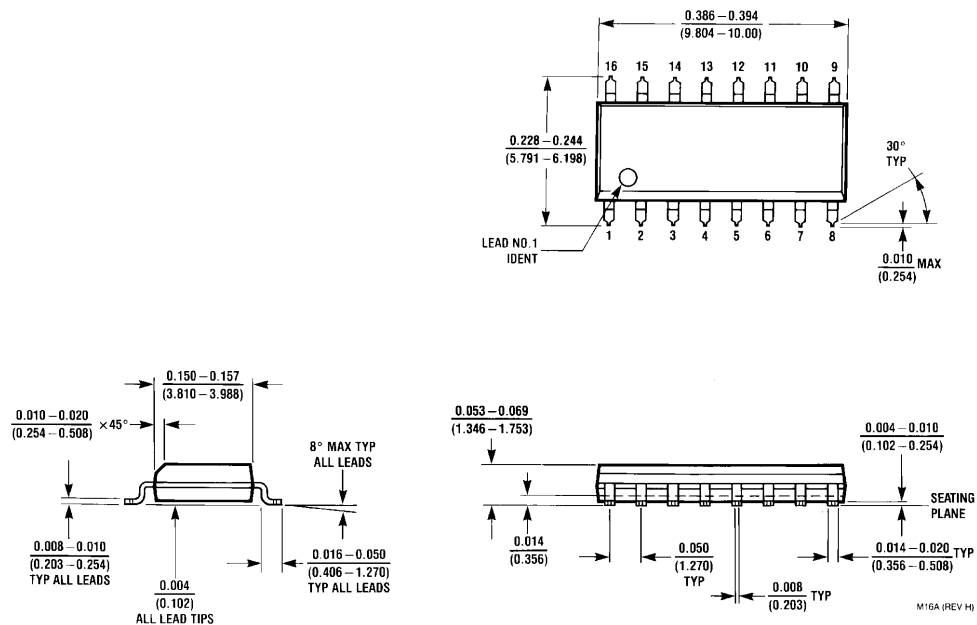
## Switching Characteristic

at  $V_{CC} = 5V$  and  $T_A = 25^\circ C$

| Symbol           | Parameter                                          | From (Input)<br>To (Output)              | R <sub>L</sub> = 2 kΩ  |     |                        |     | Units |
|------------------|----------------------------------------------------|------------------------------------------|------------------------|-----|------------------------|-----|-------|
|                  |                                                    |                                          | C <sub>L</sub> = 15 pF |     | C <sub>L</sub> = 50 pF |     |       |
|                  |                                                    |                                          | Min                    | Max | Min                    | Max |       |
| t <sub>MAX</sub> | Maximum Clock Frequency                            |                                          | 25                     |     | 20                     |     | MHz   |
| t <sub>PLH</sub> | Propagation Delay Time<br>LOW-to-HIGH Level Output | Clock to<br>Ripple Carry                 |                        | 35  |                        | 39  | ns    |
| t <sub>PHL</sub> | Propagation Delay Time<br>HIGH-to-LOW Level Output | Clock to<br>Ripple Carry                 |                        | 35  |                        | 44  | ns    |
| t <sub>PLH</sub> | Propagation Delay Time<br>LOW-to-HIGH Level Output | Clock to<br>Any Q                        |                        | 20  |                        | 24  | ns    |
| t <sub>PHL</sub> | Propagation Delay Time<br>HIGH-to-LOW Level Output | Clock to<br>Any Q                        |                        | 23  |                        | 32  | ns    |
| t <sub>PLH</sub> | Propagation Delay Time<br>LOW-to-HIGH Level Output | Enable $\overline{T}$ to<br>Ripple Carry |                        | 18  |                        | 24  | ns    |
| t <sub>PHL</sub> | Propagation Delay Time<br>HIGH-to-LOW Level Output | Enable $\overline{T}$ to<br>Ripple Carry |                        | 18  |                        | 28  | ns    |
| t <sub>PLH</sub> | Propagation Delay Time<br>LOW-to-HIGH Level Output | Up/Down to<br>Ripple Carry (Note 8)      |                        | 25  |                        | 30  | ns    |
| t <sub>PHL</sub> | Propagation Delay Time<br>HIGH-to-LOW Level Output | Up/Down to<br>Ripple Carry (Note 8)      |                        | 29  |                        | 38  | ns    |

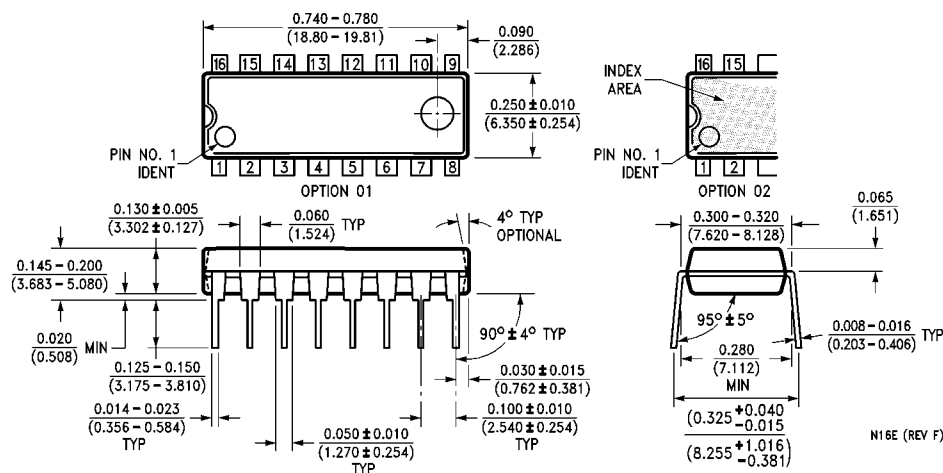
**Note 8:** The propagation delay from UP/DOWN to RIPPLE CARRY must be measured with the counter at either a minimum or a maximum count. As the logic level of the UP/DOWN input is changed, the ripple carry output will follow. If the count is minimum, the RIPPLE CARRY output transition will be in phase. If the count is maximum, the RIPPLE CARRY output will be out of phase.

# Physical Dimensions inches (millimeters) unless otherwise noted



16-Lead Small Outline Integrated Circuit (SOIC), JEDEC MS-012, 0.150 Narrow  
Package Number M16A

# Physical Dimensions inches (millimeters) unless otherwise noted (Continued)



16-Lead Plastic Dual-In-Line Package (PDIP), JEDEC MS-001, 0.300 Wide  
Package Number N16E

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